

Mid Term

COURSE: Thermodynamics of Materials

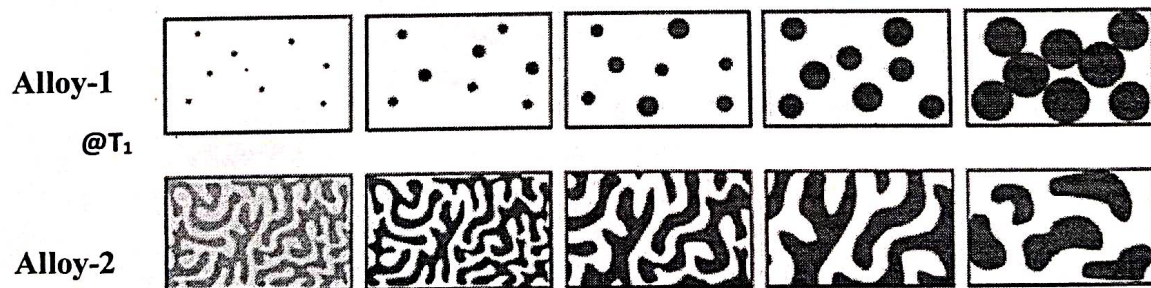
Total time: 60 mins

Answer All the Questions (Total = 30)

1. Devise a governing mathematical solution scheme to construct an Isomorphous Phase Diagram of a binary alloy system. Try to extract a mathematical expression for identifying the T_0 Curve Trajectory. Prove that this is consistent with pure metal melting event at equilibrium with constant pressure system. (Utilize the needed data base and schematics for the binary system with A and B element. Also consider $\Omega = 0$ for this binary system). *A five-point bonus marks will be awarded for applying the Newton-Raphson Method (Multivariate) to arrive at the final solution during the construction of the phase diagram.*

(7+5+3=15)

2. Observe the following sequence of microstructural evolution for two distinct alloy systems. Draw a probabilistic and schematic G vs X diagram for both the cases at a temperature T_1 . Now extrapolate the idea and draw a general T vs X curve section for both the cases. The extrapolation method should be geometrically arranged for various temperature assembly. (5)



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3. The Elements U and V have the variable interaction energy parameter. A miscibility gap exists in this binary alloy system. The interaction parameter can be expressed as $\Omega_i = \sum_{i=0}^n (1 - 2x)^i L_i$. Assume a case $n=0$ and $L_0 \neq 0$ and $L_i=0$ ($i \geq 1$). In this context, develop an expression for critical temperature T_c during a single phase to two phase transformation. Plot a schematic G vs X curve for two real number solution for $G''(x) = 0$ in interval (0,1) and only for a singular solution. (6+4=10)